

JEEVAN-AIR

Aerial Intelligence & Rescue • Software Documentation

TEAM

TEAM ZYNTAX

ORGANIZATION SPONSOR

Qualcomm Inc

CURRENT PROTOTYPE STATUS

**Software / Web GCS Prototype
(Active)**

PROBLEM STATEMENT

SIH26177

CATEGORY & THEME

Hardware / Robotics and Drones

PHYSICAL HARDWARE STATUS

Not Yet Built (Planned for Phase 4)

IMPORTANT CURRENT STATUS DISCLOSURE: The physical drone hardware, flight controller, physical cameras, and live RF transceivers DO NOT exist yet. The current system is a fully functional Ground Control Station (GCS) web prototype running mathematically simulated telemetry and sensor streams. All features are rigorously classified as [IMPLEMENTED], [SIMULATED], or [PLANNED].

2. Project Overview

JEEVAN-AIR (*Aerial Intelligence & Rescue*) is an autonomous search-and-rescue (SAR) aerial decision-support system developed by **Team ZYNTAX** for the **Smart India Hackathon 2026** under Problem Statement **SIH26177** (sponsored by **Qualcomm Inc**).

During major disasters (flash floods, earthquakes, landslides, industrial fires), ground rescue teams face extreme hazards and severe information deprivation. First responders frequently enter disaster zones without knowing whether victims are trapped or dead, where active fire or chemical hazards exist, or if roads have collapsed. JEEVAN-AIR addresses this operational void through an intelligent, closed-loop decision pipeline:

DETECT → VERIFY → ASSESS → PRIORITIZE → GUIDE

The system moves beyond simple optical object detection by fusing multi-spectral optical and thermal signatures, scoring victim urgency on an explainable 0–100 scale, ranking survivors into an operational extraction queue, and generating hazard-free ground ingress corridors for first responders.

3. Problem Statement

ID: SIH26177 • **Title:** *A deployable AI-powered autonomous drone that aids search-and-rescue operations by detecting people and hazards, thereby improving responder safety and reducing victim discovery time.*

Core Operational Deficiencies Solved:

- **Excessive Victim Discovery Latency:** Manual foot searches in large disaster zones are perilous and slow.
- **Optical False Positives:** Inanimate debris, shadows, and blankets trigger false alarms, diverting critical ground resources.
- **Lack of Urgency Prioritization:** Commercial drones deliver unranked video streams, leaving commanders unable to triage which victims require immediate extraction.
- **Danger to Ground Responders:** Ground personnel lack visibility into secondary hazards (advancing fire, toxic plumes) between their staging area and victim coordinates.

4. Proposed Solution

JEEVAN-AIR provides an integrated, deployable aerial command and decision-support architecture:

- **IMPLEMENTED Rescue Intelligence Engine:** Standalone algorithmic core ([src/services/rescueIntelligence.ts](#)) computing multi-factor risk, triage priority, and safe foot routes.
- **IMPLEMENTED Multi-Modal AI Pipeline Abstraction:** Clean interfaces for YOLOv8 person detection, radiometric LWIR thermal analysis, and spatial IoU sensor fusion ([src/ai/](#)).
- **IMPLEMENTED Tactical Ground Control Station:** 11 interactive command pages presenting real-time telemetry, GIS sector maps, alert feeds, and triage cards.
- **SIMULATED Simulation Data Adapter:** Mathematical simulation of drone flight kinematics, lawn-mower route stepping, and sensor incident injections ([SimulationDataProvider.ts](#)).
- **PLANNED Hardware Datalink Integration:** Future MAVLink 2.0 WebSocket client ([HardwareDataProvider.ts](#)) ready for physical drone connection with zero UI rewrite.

5. Current Prototype Status

Prototype Status Statement: The current JEEVAN-AIR deliverable is a comprehensive, production-quality **software Ground Control Station and decision-support prototype**. The physical UAV airframe, motors, flight controller, and physical cameras have *not yet been assembled*. All telemetry, GPS fixes, and camera streams are currently mathematically simulated.

Subsystem	Engineering Status	Implementation Details
Ground Control Station UI	IMPLEMENTED	11 operational pages in React 18, TypeScript 5.6, Tailwind CSS (Port 5180)
Rescue Intelligence Engine	IMPLEMENTED	Risk scoring (0–100), Priority Queue (#1, #2), Safe Corridor (A*), 2nd-Look
AI Sensor Fusion	IMPLEMENTED	Spatial IoU correlation (threshold ≥ 0.20), Bayesian confidence fusion
Flight & Telemetry Simulation	SIMULATED	Kinematic lawn-mower stepping, battery discharge model, Chennai GPS grid
Camera Video Feeds	SIMULATED	Canvas/SVG HUD overlays representing optical RGB, LWIR thermal, and AI overlay
Physical UAV Hardware	PLANNED	Pixhawk 6C autopilot, u-blox M9N GNSS, NVIDIA Jetson Orin Nano, FLIR Boson

6. Objectives

- 1. Minimize Victim Discovery Latency:** Accelerate aerial sector scanning through automated computer vision.
- 2. Maximize Ground Responder Safety:** Dynamically synthesize ground ingress paths avoiding lethal hazard buffers.
- 3. Eliminate False Positives:** Verify optical candidate silhouettes against biometric core body heat (35.5°C–38.2°C).
- 4. Deliver Explainable Decision Support:** Provide transparent, human-readable rationale for every risk score.
- 5. Maintain Zero-Rewrite Hardware Readiness:** Ensure the software seamlessly transitions from simulation to physical drone operations via a single configuration toggle.

7. Functional Requirements

- FR-1 (Telemetry Display):** Display drone flight mode, battery, altitude, ground speed, heading, and GPS status at 1 Hz. **IMPLEMENTED**
- FR-2 (Survivor Triage):** Score victims on a 0–100 risk scale and assign sequential priority ranks. **IMPLEMENTED**
- FR-3 (Second-Look Workflow):** Allow operators to trigger automated re-examination of ambiguous targets. **IMPLEMENTED**
- FR-4 (Safe Route Synthesis):** Calculate ground foot corridors bypassing sectors with active critical threats. **IMPLEMENTED**
- FR-5 (Dynamic Replanning):** Automatically update primary mission objective upon discovering critical victims. **IMPLEMENTED**
- FR-6 (Simulation Fallback):** Provide seamless fallback when AI models or physical sensors are unavailable. **IMPLEMENTED**

8. Non-Functional Requirements

- **NFR-1 (Performance):** Measured UI update latency $\leq 50\text{ms}$; measured AI pipeline latency tracked via `performance.now()`.
- **NFR-2 (Type Safety):** 100% strict TypeScript compilation with zero errors under `isolatedModules: true`.
- **NFR-3 (Modularity):** Complete decoupling between data adapters and presentation components via `IDataAdapter`.
- **NFR-4 (Field Resilience):** Zero external cloud runtime dependencies; capable of running on an isolated field laptop.

9. Current Software Architecture

The current software prototype implements a decoupled, event-driven Clean Architecture:

CURRENT SOFTWARE FLOW:
SimulationDataProvider → IDataAdapter Interface → Common Data Models → Rescue Intelligence → MissionContext → GCS Dashboard

The dashboard never interacts with data generation logic directly. It subscribes to the IDataAdapter contract, ensuring that switching to physical hardware requires zero UI code modifications.

10. System Architecture

Comparison between current prototype architecture and future physical drone architecture:

Layer	Current Software Prototype	Future Physical Drone Deployment
Sensing & Flight	SIMULATED Mathematical lawn-mower route stepping in SimulationDataProvider.ts	PLANNED Physical UAV + Pixhawk 6C + Sony IMX477 Optical + FLIR Boson+ 320 LWIR
Edge Computing	SIMULATED Host browser execution	PLANNED NVIDIA Jetson Orin Nano 8GB running TensorRT YOLOv8
Datalink Bridge	IMPLEMENTED In-memory event bus (TypeScript Set<()=>void>)	PLANNED JeevanAir Edge Bridge (Python/FastAPI WebSocket + 2.4GHz RF)
Adapter Ingest	IMPLEMENTED SimulationDataProvider.ts	PLANNED HardwareDataProvider.ts (WebSocket client)
Intelligence Core	IMPLEMENTED rescueIntelligence.ts (Risk, Priority, Safe Route)	IMPLEMENTED Identical code reused without alteration
Command Center	IMPLEMENTED React 18 GCS Dashboard (11 Tactical Pages)	IMPLEMENTED Identical UI reused without alteration

11. Data Flow

MULTI-MODAL DECISION DATA FLOW:
Optical RGB Frame + Thermal LWIR Frame → YOLOv8 Detection + Radiometric Analysis
→ MultiModalFusionEngine (Spatial IoU Correlation ≥ 0.20)
→ FusedAIDetection (Confidence, Biometric Heat Verification)
→ assessSurvivorRisk() [5-Factor Multi-Factor Score 0-100]
→ prioritizeSurvivors() [Sequential Urgency Ranking #1, #2...]
→ calculateSafeRoute() [A* Ground Hazard-Bypass Corridor]
→ MissionContext State Update → Reactive UI Re-render

12. Software Modules

Directory / File	Layer	Primary Role	Status
src/types/common.ts	Domain	Canonical data models (Telemetry, Survivor, Hazard, Alert, Route)	IMPLEMENTED
src/adapters/IDataAdapter.ts	Adapter	Abstract interface contract for all telemetry providers	IMPLEMENTED
src/adapters/SimulationDataProvider.ts	Adapter	Simulated flight kinematics, lawn-mower route, incident generator	SIMULATED
src/adapters/HardwareDataProvider.ts	Adapter	Future WebSocket MAVLink client stub	PLANNED
src/adapters/providerFactory.ts	Adapter	Factory reading SIMULATION_MODE for single-switch selection	IMPLEMENTED
src/services/rescueIntelligence.ts	Intelligence	Risk scoring, priority sorting, A* safe routing, second-look resolution	IMPLEMENTED
src/ai/AIPipelineService.ts	AI / Vision	Multi-modal pipeline orchestrator and benchmark tracking	IMPLEMENTED
src/ai/MultiModalFusionEngine.ts	AI / Vision	Spatial IoU correlation and Bayesian confidence booster	IMPLEMENTED
src/hardware/config.ts	Hardware Prep	SIMULATION_MODE toggle and 18-item readiness checklist	IMPLEMENTED

13. Survivor Detection

IMPLEMENTED Abstraction · **SIMULATED** Frames

Combines optical silhouette detection with radiometric LWIR thermal verification:

- **Optical Detection (`yoloVisionProvider.ts`):** Detects human silhouettes, outputting bounding box coordinates (x, y, w, h) and confidence.
- **Thermal Verification (`ThermalProcessingProvider.ts`):** Evaluates radiometric heat matrix to confirm human core temperatures $(35.5^{\circ}\text{C} \text{--} 38.2^{\circ}\text{C})$.
- **IoU Sensor Fusion (`MultiModalFusionEngine.ts`):** Correlates optical and thermal detections spatially with an IoU threshold of 0.20. Boosts confidence on dual confirmation, discounts confidence on optical-only detections.

14. Hazard Detection

IMPLEMENTED Interface · **SIMULATED** Incidents

Tracks 8 operational disaster hazard categories via `HazardDetectionProvider.ts`:

- **Fire:** Thermal combustion detection ($T > 60^{\circ}\text{C}$), affected radius 50 m.
- **Flood / Flooded Area:** Specular reflectance and water boundary edge indices.
- **Smoke:** Atmospheric contrast degradation and aerosol dispersion.
- **Debris & Damaged Structure:** Surface texture roughness and geometric collapse profiles.
- **Chemical Hazard:** Plume dispersion tracking and thermal vapor contrast.

15. Rescue Intelligence Core

IMPLEMENTED Standalone Module (`src/services/rescueIntelligence.ts`)

Transforms raw sensing data into actionable field directives. Implements pure functional logic independent of UI frameworks, validated by 24 automated unit tests.

16. Risk Assessment

IMPLEMENTED Scoring Algorithm

Computes an explainable risk score (0–100) across five weighted factors:

Evaluation Factor	Weight	Scoring Rules
Movement Status	0–30 pts	<code>NO_MOVEMENT</code> / <code>STATIC</code> : +28 <code>UNKNOWN</code> : +15 <code>MOVEMENT_DETECTED</code> : +10
Estimated Condition	0–25 pts	<code>CRITICAL</code> : +25 <code>UNCERTAIN</code> : +18 <code>POSSIBLE</code> : +14 <code>STABLE</code> : +8
Thermal Confirmation	0–15 pts	Confirmed biometric heat ($35.5^{\circ}\text{C} \text{--} 38.2^{\circ}\text{C}$): +15 Unconfirmed: +5
Hazard Proximity	0–20 pts	Within buffer of Critical hazard: +20 High hazard: +15 Distance < 50m: +10
Terrain Accessibility	0–10 pts	Flooded / impassable terrain: +10 Debris obstruction: +5 Clear: +2

Severity Mapping: CRITICAL (≥ 75), HIGH (50–74), MEDIUM (25–49), LOW (0–24).

17. Rescue Prioritization

IMPLEMENTED Priority Queue

Sorts survivors into an actionable queue:

1. Filters out rejected false positives.
2. Sorts primarily by descending `riskScore`.
3. Resolves ties via sensor `confidence`.
4. Assigns ordinal ranks (`Priority #1`, `Priority #2` ...) rendered on dashboard badges.

18. Second-Look Verification

IMPLEMENTED Verification Workflow

Ambiguous detections (low confidence, optical-only) are placed in `POSSIBLE` status. Clicking `REQUEST SECOND LOOK` transitions the target to `UNDER_REVIEW` and simulates multi-spectral re-examination: confirming genuine survivors to `VERIFIED` (confidence elevated $\geq 88\%$) or rejecting false-positive debris to `REJECTED`.

19. Safe-Access Guidance

IMPLEMENTED A* Ingress Pathfinding

Fundamental Distinction: Drone Flight Route $\neq$ Ground Responder Route. The drone flies straight over rivers and fires, but ground rescue teams require safe paths on foot.

The pathfinding engine evaluates a 3×3 sector graph, marks sectors with active critical hazards as impassable obstacles, and computes the shortest hazard-free corridor from base (A1) to the victim, including walking travel time at 1.1 m/s .

20. Mission Replanning

IMPLEMENTED Dynamic Objective Engine

Automatically retargets the drone from broad sweep mode to localized target monitoring upon discovering critical victims or emerging fires, updating the mission banner and timeline.

21. Dashboard Architecture

IMPLEMENTED 11 Operational Pages

Dashboard (command center), Mission Control (flight buttons & injections), Live Search (multi-spectral viewport), Detections (ledger), Hazards (threat tracker), Search Map (GIS grid), Alerts (notification center), Mission History (logs), System Status (health gauges), System Architecture (diagrams), and HW Integration (readiness checklist).

22. Simulation Architecture

SIMULATED Kinematic Simulation Engine

Generates realistic drone position along a 10-waypoint lawn-mower route across sectors A1–C3, decrements battery from 98%, models 3D GPS lock at 13.0827°N, 80.2707°E, and provides scenario injection controls.

23. AI Architecture

IMPLEMENTED Measured Performance Pipeline

Measures actual execution latency via `performance.now()`. Features simulation fallback mode badged in UI when models are offline. Zero fabricated benchmark numbers.

24. Connectivity

SIMULATED Current State · **PLANNED** Hardware Bridge

Current prototype models three communication states: `SIMULATED` — `CONNECTED`, `DEGRADED`, and `DISCONNECTED`. Future architecture specifies 2.4GHz RF MAVLink telemetry and 4G/LTE failover.

25. Security

PLANNED Deployment Architecture

MAVLink 2.0 packet signing via SHA-256 HMAC tokens, role-based access control (Commander vs. Responder), hardware geofencing with automated RTL fail-safe, and isolated local RF/Wi-Fi mesh operation.

26. Error Handling

IMPLEMENTED Fault-Tolerant Defenses

Hardware provider returns safe fallback telemetry when disconnected; A* pathfinder returns `BLOCKED` with advisories rather than crashing when access is cut off; AI pipeline falls back gracefully to synthetic generation on model failure.

27. Testing & Verification

IMPLEMENTED 85 Automated Assertions (100% Pass Rate)

- **Phase 2 Intelligence Suite (`rescueIntelligence.test.ts`)**: 24 / 24 PASSED (Risk scoring, priority sorting, A* routing, second-look resolution, timeline events).

- **Phase 3 AI Pipeline Suite (`aiPipeline.test.ts`):** 61 / 61 PASSED (YOLO optical, thermal radiometric, hazard detection, IoU fusion, fallback mode, performance measurement).
- **Build Verification:** `npm run build` passes cleanly in 1.4s with zero TypeScript warnings.

28. Current Limitations

1. **Physical Drone Absent:** Airframe, motors, ESCs, and battery pack are not physically constructed.
2. **Hardware Sensors Absent:** Optical and thermal cameras are simulated; no physical video feeds.
3. **No Live MAVLink Datalink:** Telemetry is generated via mathematical formulas in software.
4. **Coarse 3x3 Grid:** Ground routing operates across 9 discrete sectors rather than continuous DEM terrain maps.
5. **Client-Side Execution:** AI inference runs on the browser CPU rather than an onboard NVIDIA Jetson.

29. Future Hardware Integration

PLANNED

Zero-Rewrite Hot-Swap Architecture

The system is pre-engineered for physical hardware. When the physical drone is constructed:

1. Deploy JeevanAir Edge Bridge (Python/FastAPI) on NVIDIA Jetson Orin Nano connected to Pixhawk via UART.
2. In `src/hardware/config.ts`, set `SIMULATION_MODE = false` and configure WebSocket URL.
3. `providerFactory.ts` automatically switches to `HardwareDataProvider` . **Zero UI code modified.**

30. Future Scope

- **Continuous 3D Topographical Routing:** High-resolution LiDAR Digital Elevation Models (DEM) with slope hazard modeling.
- **Multi-UAV Swarm Mesh:** Decentralized autonomous coordination across multiple search drones.
- **Emergency Payload Drop:** Actuated servo release for medical kits, water purification, and emergency radios.

31. Technology Stack

Layer	Technology	Version	Role
Language	TypeScript	5.6.3	Strictly typed domain models, interfaces, and algorithms
UI Framework	React	18.3.1	Component tree, reactive context state, tactical layout
Styling	Tailwind CSS	3.4.14	Tactical dark-theme utility styling and responsive grid
Build & Dev	Vite	5.4.8	Fast HMR dev server (Port 5180) and Rollup production bundler
Icons	Lucide React	1.16.0	Tactical radar, drone, alert, hazard iconography
Testing	tsx	Native	TypeScript test runner executing 85 automated assertions

32. Project Structure

```
jeevan-air/
├── docs/                                # Technical specifications & documentation
│   ├── JEEVAN-AIR-Software-Documentation.md
│   ├── JEEVAN-AIR-Prototype.md
│   └── ER-Diagram.md
├── src/
│   ├── adapters/                        # Data Provider Layer
│   │   ├── IDataAdapter.ts             # Abstract provider contract
│   │   ├── SimulationDataProvider.ts    # [SIMULATED] Concrete flight & sensor simulation
│   │   ├── HardwareDataProvider.ts     # [PLANNED] Future MAVLink WebSocket client
│   │   └── providerFactory.ts          # Hot-swap provider factory selector
│   ├── ai/                             # Multi-Modal AI & Vision Module
│   │   ├── AIPipelineService.ts        # Central orchestrator with performance.now()
│   │   ├── YoloVisionProvider.ts       # Optical YOLOv8 person detection abstraction
│   │   ├── ThermalProcessingProvider.ts # Radiometric LWIR thermal analyzer
│   │   ├── HazardDetectionProvider.ts  # Extensible hazard classifier interface
│   │   └── MultiModalFusionEngine.ts    # Spatial IoU sensor fusion engine
│   ├── components/                     # Reusable Tactical UI Components
│   │   ├── ai/AIInferencePanel.tsx     # Live inference diagnostics panel
│   │   ├── intelligence/               # Rescue Intelligence components
│   │   │   ├── DecisionTimeline.tsx    # Chronological audit log
│   │   │   ├── MissionObjectiveBanner.tsx # Dynamic replanning banner
│   │   │   └── RescuePriorityPanel.tsx  # Priority queue, explainable cards, routes
│   │   ├── map/SearchMap.tsx           # Tactical 3x3 sector GIS map
│   │   └── camera/CameraFeed.tsx       # Multi-spectral camera viewport
│   ├── hardware/                       # Phase 4 Hardware Integration Layer
│   │   ├── config.ts                  # SIMULATION_MODE toggle & readiness checklist
│   │   └── types.ts                   # Complete 12-schema MAVLink telemetry types
│   ├── services/
│   │   └── rescueIntelligence.ts       # Pure algorithms: Risk, Priority, A* Route
│   ├── tests/
│   │   ├── rescueIntelligence.test.ts  # 24 Phase 2 intelligence assertions
│   │   └── aiPipeline.test.ts          # 61 Phase 3 AI pipeline assertions
│   ├── types/common.ts                 # Canonical Common Data Model interfaces
│   └── package.json                    # Scripts & dependencies
```

33. Installation

```
# 1. Clone repository
git clone https://github.com/Srisharadhakrishnan/SIH26177.git
cd SIH26177

# 2. Install dependencies
npm install

# 3. Verify TypeScript and production build
npm run build
```

34. Running the Project

```
# Start local dev server (configured on port 5180 to avoid collisions)
npm run dev
# Open browser at: http://localhost:5180/

# Run complete test suite (85 assertions)
npm run test:all
```

35. Conclusion

The **JEEVAN-AIR** software prototype delivers a verified, highly capable Ground Control Station and aerial decision-support system. By solving the critical gap between raw aerial vision and ground responder action, the system demonstrates how AI-powered drones can drastically reduce victim discovery time and protect responder lives during disasters. Engineered with clean separation of concerns and single-switch hardware readiness, JEEVAN-AIR is ready for SIH evaluation, technical review, and physical hardware deployment.

Appendix: Comprehensive Feature Classification Matrix

Subsystem / Feature	Category	Classification	Technical Scope & Implementation
Common Data Model	Architecture	IMPLEMENTED	Strongly typed domain models in <code>src/types/common.ts</code>
Data Adapter Contract	Architecture	IMPLEMENTED	<code>IDataAdapter</code> abstract interface decoupling simulation from UI
Provider Factory	Architecture	IMPLEMENTED	Single-switch hot-swap factory reading <code>SIMULATION_MODE</code>
Explainable Risk Scoring	Intelligence	IMPLEMENTED	5-factor scoring (0–100) with explainable rationale strings
Rescue Priority Queue	Intelligence	IMPLEMENTED	Active triage sorting (#1, #2...) with tie-breakers
Safe-Access Ground Routing	Intelligence	IMPLEMENTED	A* pathfinding on 3×3 sector grid avoiding active hazard zones
Second-Look Verification	Intelligence	IMPLEMENTED	Ambiguity resolution workflow eliminating false-positive debris
Spatial IoU Sensor Fusion	AI / Vision	IMPLEMENTED	Multi-modal bounding box correlation and confidence boost
Measured Latency Tracking	AI / Vision	IMPLEMENTED	Actual execution benchmarking via <code>performance.now()</code>
AI Simulation Fallback	AI / Vision	IMPLEMENTED	Seamless fallback mode badged in UI when models are offline
Tactical GCS Dashboard	User Interface	IMPLEMENTED	11 operational pages in dark tactical theme on Port 5180
Drone Flight Kinematics	Simulation	SIMULATED	Stepped lawn-mower route across sectors A1–C3 at 3.5 m/s
Drone Telemetry Gauges	Simulation	SIMULATED	Altitude (32m AGL), speed, heading, and battery discharge curve
GPS Satellite Positioning	Simulation	SIMULATED	Mathematical 3D fix centered on Chennai disaster coordinates
Camera Video Streams	Simulation	SIMULATED	Canvas/SVG HUD overlays simulating optical, thermal, and AI overlay
Physical UAV Airframe	Hardware	PLANNED	Quadcopter frame, brushless motors, ESCs, and LiPo power system
Physical Autopilot	Hardware	PLANNED	Pixhawk 6C running ArduCopter / PX4 with MAVLink 2.0
Physical Cameras	Hardware	PLANNED	Sony IMX477 optical camera and FLIR Boson+ 320 LWIR thermal camera
Onboard Edge Computer	Hardware	PLANNED	NVIDIA Jetson Orin Nano running TensorRT YOLOv8
Edge WebSocket Bridge	Hardware	PLANNED	Python/FastAPI service broadcasting <code>HardwareTelemetryPacket</code> at 10 Hz